

Education

- Dr. B. R. Ambedkar National Institute of Technology, Jalandhar**
Jalandhar, Punjab
B.Tech in Computer Science and Engineering; CGPA: 9.04 / 10
Aug 2024 - May 2028
- Sarvottam Academy, Barnala** | *Senior Secondary (Class XII)* — **88%**
2024
- YS Public School, Barnala** | *Secondary (Class X)* — **95%**
2022

Projects

- WinDrop** | *Full-Stack Developer* | *MERN Stack, C++, Socket.io, TCP Sockets*
April 2026
[GitHub Code](#)
 - Hybrid Core:** Engineered a P2P network pairing a React.js interface with a high-performance C++ transmission engine via a responsive Node.js/Express orchestration backend.
 - Transport Engine:** Programmed low-latency transport layer workflows natively with C++ TCP Sockets, optimizing memory allocation for massive parallel file streams.
 - Peer Handshakes:** Integrated real-time client signaling and dynamic network discovery using Socket.io, maintaining file transfer metadata safely within MongoDB.
- Lumira** | *Full-Stack Developer* | *MERN Stack, React Three Fiber, Ollama, LLMs*
2026
[GitHub Code](#)
 - AI Socratic Model:** Formulated an intelligent inquiry-driven educational framework leveraging persistent context storage to guide students to answers fluidly.
 - Local LLM Stack:** Developed secure local inference pipelines using Ollama hooked into a structural Node.js framework backed by robust MongoDB state tracking.
 - 3D Interaction Interface:** Constructed an immersive visual layout using React Three Fiber, translating conversational pipeline phases into smooth avatar animations.
- AgriLink** | *Full-Stack Developer* | *React.js, Gemini AI, Web Speech API, Node.js*
Jan 2026 - Present
[GitHub Code](#) | [Live Demo](#)
 - Voice Accessibility:** Integrated Web Speech API and Gemini AI inside a React build to deliver high-utility agricultural market intelligence via spoken commands.
 - Fintech Architecture:** Created optimized backend transactional routes for trading workflows including custom Micro-Pooling and Reverse Bidding schemes.

Achievements

- Competitive Programming:** Achieved **Knight** badge on LeetCode (1850 rating) and actively compete in global Codeforces contests to continuously hone algorithmic skills under time constraints.
- Data Structures and Algorithms:** Solved **600+** problems on LeetCode, building robust problem-solving skills and a strong foundation in complex concepts including Trees, Graphs, and Dynamic Programming.
- Hackathons & Competitions:** Secured Second Runner-Up at Innovation with TRAE Hackathon NITJ. Participated in The Big Code 2026 and developed an accessibility website for individuals with memory impairments (Mar 2026).
- Google Solution Challenge 2026 (Team FarmHers):** Engineered a voice-first agri-fintech ecosystem focused on agricultural sustainability.

Skills

- Languages & Core Concepts:** C, C++, Python, Java, JavaScript | DSA, OOP, DBMS, Computer Networks
- Frontend & Web:** React, HTML, CSS, Bootstrap
- Backend & Databases:** Node.js, Express.js, MongoDB, Supabase, REST APIs
- Tools & Platforms:** Git, GitHub, VS Code, Netlify